

Optic Disc Segmentation Based on Mask R-CNN in Retinal Fundus Images

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Abstract— An optic disc is an object on the retina of the eye that has the characteristics of being brightly colored and round. Optic disc segmentation is the most common step taken before processing a retinal fundus image. The bright characteristics of the optic disc often interfere with the detection of other objects in the retinal fundus image. Therefore, the optic disc is the first step before processing the fundus image of the retina. With the help of digital image processing will help in the removal of the optic disc from the fundus image of the retina. Many methods can be used in optic disc segmentation, one of which is the deep learning method. The deep learning method chosen is Mask R-CNN to produce a mask from the results of object detection on the retinal fundus image. There are 3 stages in the segmentation process using the Mask R-CNN. First, the data used in the training process will be labeled. there is 1 label given, namely optic disc. Then the model is trained using the resnet50 backbone architecture and finally, the model will be evaluated. To evaluate the results obtained from the two methods, it uses Intersection over Union (IoU) by comparing directly the results of prediction and ground truth. The data used is an IDRiD dataset containing retinal fundus images taken from eye clinics across India. As the result, Mask R-CNN can segment the optical disc with an IoU value of 0.843. it is hoped that the results of this research can help the process in processing retinal fundus images in the future.

Keywords—diabetic retinopathy, optic disc, segmentation, Mask R-CNN, IoU

I. INTRODUCTION

Diabetes mellitus is a chronic disease suffered by many people around the world. One of the diseases caused by diabetes is diabetic retinopathy [13]. Diabetic retinopathy is the most common cause of blindness or visual impairment in people of work-ready age. The International of Ophthalmology reports that 1 in 3 diabetic patients is affected by diabetic retinopathy. Diabetic retinopathy can be diagnosed by looking at one or more symptoms such as microaneurysms, hemorrhages, hard exudates, and soft exudates on the retina [9]. The initial diagnosis of diabetic retinopathy can be seen from the presence of disturbances in the patient's vision.

The optic disc is one of the main anatomies of the retina that is circular and the brightest anatomy in the retina. The optic disc or nerve center of the eye is the place where the nerve enters the eye area. The optic disc is also the meeting place of all the nerves present in the eye [11]. The optic disc consists of 1.2 million ganglion cell axons that pass through the retina to the eye sending visual information from the eye to the brain. To be able to increase success in segmentation on retinal images of microaneurysms, hemorrhages, hard

exudates, and soft exudates requires a normal anatomical separation process such as an optic disc.

Optical disc segmentation is the most important and common step taken when processing retinal images [1]. Segmentation of the optic disc helps in diagnosing diseases of the eye such as diabetic retinopathy and glaucoma. Separation of optic discs in diagnosing diabetic retinopathy needs to be done to avoid errors in detecting symptoms of diabetic retinopathy caused by similarities in both color, contrast, and harmony of the optic disc with symptoms of diabetic retinopathy.

There are several related studies on the segmentation of optic discs in retinal fundus images Elahi *et al* [3]. Segmentation and localization of optic disc and Fovea from diabetic retinopathy image received MIoU values of 0.845, 0.901, 0.933, and 0.920 for IDRiD, RIMONE, DRISHTI-GS, and DRIVE. Related research was also conducted by Hermawati and Safii [5] to Detect Lung Cancer Malignancy in CT Scans using the Region-based Convolutional Neural Network (Mask R-CNN) Mask Method. This study discussed the performance produced by the R-CNN Mask method for segmenting tumor areas in the CT scan module of the R-CNN Mask method, successfully segmenting tumor areas with high performance with an average value of precision of 0.852, sensitivity of 0.958, specificity of 0.82, and Dice Similarity of 0.894.

Several methods can be used in segmenting objects on images such as deep learning methods. The deep learning method used is Mask R-CNN. The R-CNN mask is a simple method in the training model process but can produce good output results by detecting every object in the image, the r-cnn mask can also be used for various object detection problems. Mask R-CNN has the concept of imitating the workings of the human eye. By imitating this way of working, R-CNN Mask can recognize its objects based solely on images. The R-CNN mask is an addition to the previous method of Fast R-CNN [6]. It is said to be an addition to the previous method because the Mask R-CNN method produces 3 outputs, namely the bounding box, class, and mask of the class. In training the previous Mask R-CNN model the data needs to be labeled so that the system can learn to recognize objects from those labels.

The purpose of this study is a system that can segment the optic disc on the retinal fundus image quickly using the Mask R-CNN method. Segmenting the optic disc on the retinal fundus image can help the image processing research process related to the retinal fundus image.

II. RELATED WORKS

Based on previous research on optic disc segmentation with the digital image processing process, there are several studies that are relevant to the research that will be carried out including the following:

The research conducted by Elahi, *et al* [3] in a study entitled "Segmentation and localization of optic disc and Fovea from diabetic retinopathy image" can segment optic discs well on several data sets. The IoU values obtained were 0.845, 0.901, 0.933, and 0.920 for IDRiD, RIMONE, DRISHTI-GS, and DRIVE. Another related research was conducted by (Aswathy & Kumar, 2021) [2] in a study entitled "Diabetic Retinopathy Detection Using Semantic Segmentation and Optic Disc Localization". This study produced good optical disc segmentation and localization results by obtaining an accuracy value of 97.3%. Other related research on optic disc segmentation is quite common in [9], [10], [12], [14]

Based on previous research, this study will focus on optical disc segmentation using deep learning methods with the R-CNN Mask architecture. It is hoped that the results of this study will get better results than previous studies.

III. DATASET

Retinal fundus image set data is processed through the Indian Diabetic Retinopathy Image Dataset (IDRiD) [8] website. The source of the dataset comes from an Indian population obtained through an eye clinic located in Nanded, India. The data taken is a photo of the eyes of patients with diabetes. The data was collected from 2007 to 2017 and the quality has been verified by experts. Taking photos of the retina of the eye using the Kowa VX-10 α digital fundus camera with a field of view (FOV) of 50 $^\circ$ and an eye spacing with a lens 39 mm long and equipped with a xenon flash. The resulting photo has a resolution of 4288 x 2848 pixels and is stored in jpg form with a size per file of 800kb. Fig. 1 displays an example of an IDRiD dataset with a searchable object on this dataset.

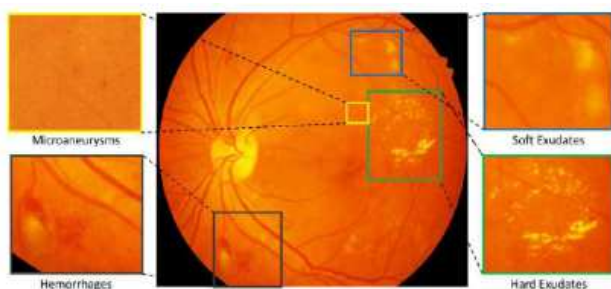


Fig. 1. Fundus image of the retina infected with diabetic retinopathy. Enlarged parts are microaneurysms, soft exudates, hemorrhages, and hard exudates

The Indian Diabetic Retinopathy Image Dataset (IDRiD) has a total of 81 original images and 81 ground truth image of optic disc. The original image is divided into 2 groups, namely the training set with a total of 54 images and the testing set with a number of images 27. The ground truth image is divided into 2 groups just like the original, namely training and testing sets which are further divided into 5 parts, namely optic discs, hard exudates, soft exudates, microaneurysms, and hemorrhages with a total of 54 data

from the training set section and 27 for testing. Fig. 2 displays a sample from IDRiD dataset.

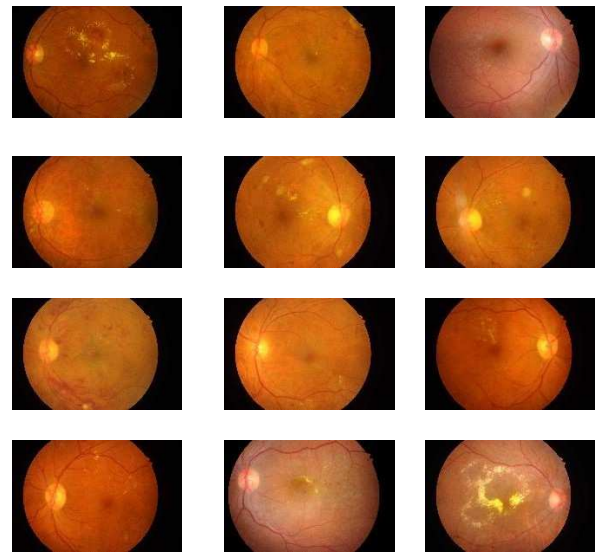


Fig. 2. Sample of the dataset

The making of ground truth is done by master-level students using special software developed by ADCIS which is specifically used for the Anastasi process. The contour of the retinal fundus image is made manually. Then the object sought was given a sign in the form of a line that surrounds and is validated for its correctness by 2 ophthalmologists, namely Vivek Sahasrabudhe, an ophthalmology specialist from the Department of Ophthalmology, Shankarrao Chavan Government Medical College, and Dr. Girish Deshmukh, an Ophthalmologist from the Eye Clinic, Sushrusha Hospital. The result of ground truth is displayed in Fig. 3.

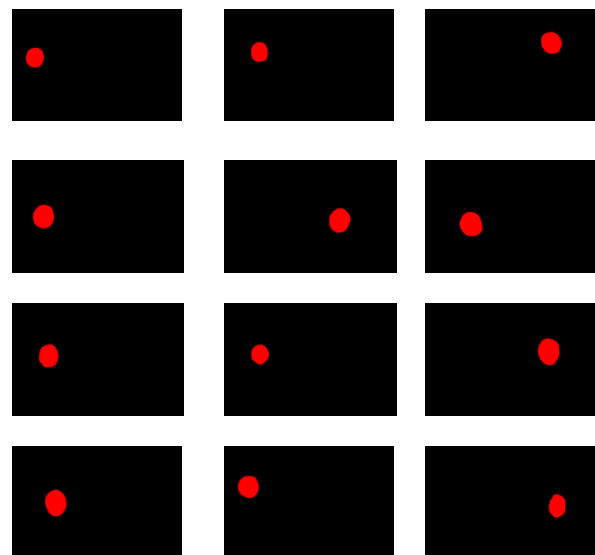


Fig. 3. Sample of Ground Truth

IV. MASK R-CNN ARCHITECTURE

The method used in the study was Mask R-CNN. Mask R-CNN is a branch or continuation of the previous method, namely Faster R-CNN. The output of the R-CNN Mask is a mask from the detection results of the Faster R-CNN architecture. The R-CNN mask was developed in 2017 by Kaiming He.

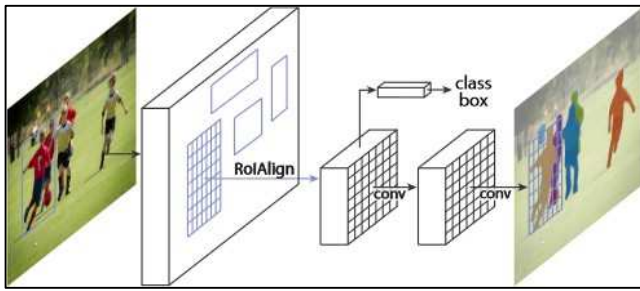


Fig. 4. Illustration of Mask R-CNN structure

Fig. 4 shows how the Mask R-CNN architecture works. There are 3 stages in the segmentation process using the R-CNN Mask, the first stage is the Region Proposal Network (RPN) stage to get candidates for object bounding boxes, then the 2nd stage is to determine the class of each object by extracting features using RoIAlign. [5]. In addition to producing the results of class prediction and bounding boxes, its objections in the 2nd stage of the R-CNN Mask also produce masks of each object. Details of the parameters used during the model training process using the Mask R-CNN method can be seen in Table I.

TABLE I. DETAIL PARAMETER OF MASK R-CNN

Parameters	Description
Batch Size	4
Backbone	Resnet50
Learning Rate	0.001
Epoch	20
Step per Epoch	600
Validation step	100
Image Min Dimension	512
Image Max Dimension	512

V. METHODOLOGY

Fig. 5 shows the working stages of the Mask R-CNN. The initial stage of the Mask R-CNN method is to label the data, the data that has been labeled will be a role in the input for the process. In the training model process using the Resnet50 architecture backend. After getting the trained model to see what the prediction results are with ground truth. The prediction results will be evaluated using the IoU method.

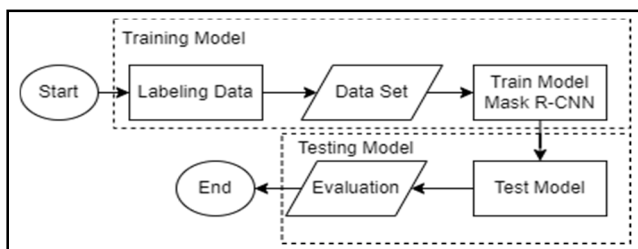


Fig. 5. Workflow of Mask R-CNN

To evaluate the value of optical disc segmentation will use the method intersection over union (IoU) method. IoU is a metric evaluation method that is used to measure the accuracy of segmentation results by looking for the slice value and the combination of the prediction result with the ground truth, then the slice value will be divided by the combined value resulting in the IoU value. which are expressed by (1).

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (1)$$

VI. RESULT AND DISCUSSION

This section will describe the result of the application from the workflow in the method section and then evaluate the method using Intersection over Union (IoU).

The data that will be used in the training process uses 54 data according to the provisions of the dataset used. The data trained for the model of the model is first labeled. Data labeling is done manually using the anataste website called makesense.ai. The number of data labeled is 54 images with 1 optical disc label. the process of data labeling can be seen in Fig. 6.

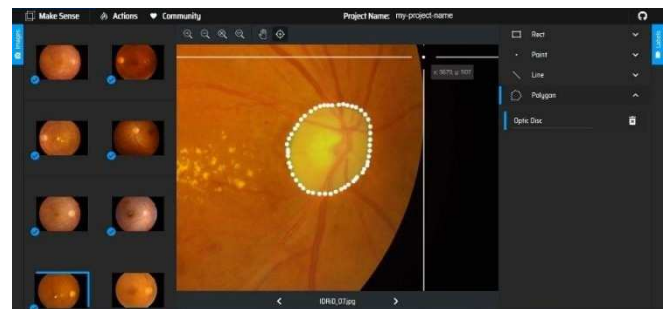


Fig. 6. Data Labeling Process

The distribution of data to be used in the training process uses 54 data divided into 48 data for training and 6 data for testing and there is 1 label class. The data is divided randomly. Data Training has carried out as many as 20 epochs with a step a per epoch of 600 and a validation step of 100. The number of data used for the training model was 54 data which was divided into 48 images for training and 6 images for validation. The results of the Mask R-CNN model training can be seen in Fig. 7. The loss value obtained on the 20th epoch was 0.0149 with a validation loss of 0.0128.

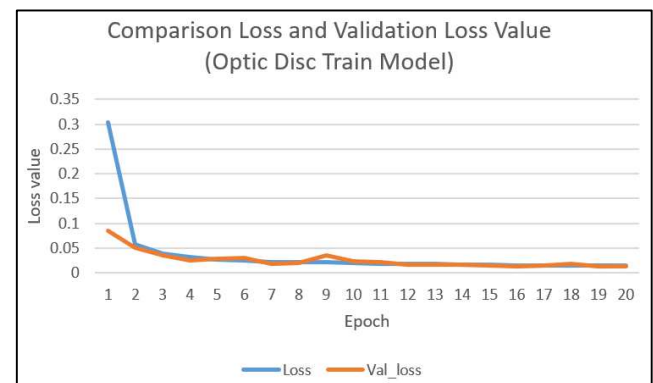


Fig. 7. Model Loss and val loss of training

After getting the mode from the training results, the next step is to evaluate the model using 27 testing images in accordance with the provisions in the dataset used. The evaluation value is obtained using the IoU method by comparing the prediction results with the ground truth. The IoU value obtained after testing at 27 is 0.845. Compared to previous research, the results obtained are fairly good because they get an mIoU value greater than other methods

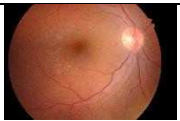
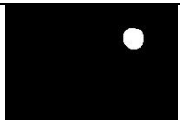
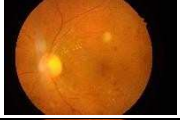
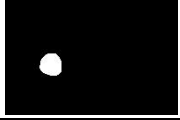
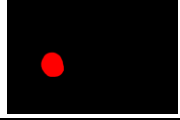
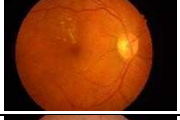
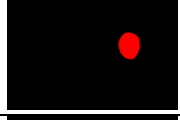
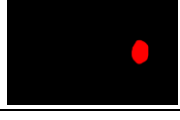
and get a relatively similar value to the DRNet method that presented in Table II.

TABLE II. COMPARISON RESULT FOR MARK-CNN AND OTHER METHODS

Author	Different Methods	mIoU
Porwal P (2018)	L ₀ -GM	0.712
Furtado P (2020)	FCN	0.680
Furtado P (2020)	DeepLab	0.768
Furtado P (2020)	Unet	0.173
Elahi (2021)	DRNet	0.845
This work	Mask R-CNN	0.845

The results of the segmentation can be seen in Table III, the results obtained are already good with only a few results that have a non-round shape.

TABLE III. COMPARISON BETWEEN PREDICTED RESULTS AND GROUND TRUTHS

Original Image	Prediction	Ground truth
		
		
		
		

VII. CONCLUSION

In this paper, we implement the Mask R-CNN method to segment the optic disc on the fundus image of the retina of patients with diabetic retinopathy. The method we use uses a total of 81 images which are divided into 54 images for the model training process and 27 testing images for model evaluation. The training model has carried out as many as 20 epochs with a loss of 0.0149 and a validation loss of 0.0128. After getting the model from the training model, then the model was evaluated using 27 images with the IoU method. The IoU value from the model evaluation results is 0.843.

From these results, the model produces fairly good results because get better results compared to previously conducted research. In the next study, we will use this result as a base for segmenting the other diabetic retinopathy symptoms, namely exudates, hemorrhages, and microaneurysms.

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